

AI-DND

INVENTORY_SYSTEM.md

Inventory and Equipment System

Version 0.1

1. DESIGN PHILOSOPHY

Inventory is not an infinite container.

The player carries physical objects.

Every item has:

- weight;
- size;
- condition;
- value;
- origin;
- ownership.

Inventory management should create meaningful decisions.

2. INVENTORY TYPES

The player possesses several inventories.

Body Equipment

Equipped items.

Backpack

Main storage.

Belt

Quick-access items.

Pockets

Small items only.

Mount Inventory

Animal storage.

Home Storage

Permanent storage.

3. EQUIPMENT SLOTS

Character equipment slots:

- head;
 - face;
 - neck;
 - torso;
 - cloak;
 - gloves;
 - belt;
 - pants;
 - boots;
 - left ring;
 - right ring;
 - amulet;
 - primary weapon;
 - secondary weapon;
 - shield;
 - ranged weapon;
 - ammunition.
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4. ITEM CATEGORIES

Weapons

Examples:

- swords;
 - axes;
 - maces;
 - daggers;
 - spears;
 - bows;
 - crossbows;
 - staffs.
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Armor

Examples:

- cloth;
 - leather;
 - chainmail;
 - plate.
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Accessories

Examples:

- rings;
 - amulets;
 - talismans.
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Consumables

Examples:

- food;
- potions;
- medicine;
- alcohol.

Materials

Examples:

- ore;
- wood;
- leather;
- herbs;
- crystals.

Documents

Examples:

- letters;
- maps;
- books;
- contracts.

Quest Items

Examples:

- relics;
- keys;
- artifacts.

5. WEIGHT SYSTEM

Every item has weight.

Examples:

Sword: 2.5 kg

Apple: 0.2 kg

Armor: 18 kg

Weight affects:

- stamina;
 - travel speed;
 - combat;
 - fatigue.
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6. VOLUME SYSTEM

Items occupy physical space.

Examples:

Ring: very small.

Sword: large.

Armor: very large.

7. CARRYING CAPACITY

Formula:

Strength

+ Training

+ Equipment

determines carrying capacity.

8. OVERLOAD

Effects of overload:

- slower movement;
 - increased fatigue;
 - combat penalties;
 - injury risk.
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9. ITEM QUALITY

Quality levels:

Broken ↓ Poor ↓ Common ↓ Good ↓ Excellent ↓ Masterwork ↓ Legendary

10. ITEM CONDITION

Items degrade.

Examples:

- rust;
- cracks;
- dullness;
- wear.

Condition affects performance.

11. DURABILITY

Weapons:

- lose sharpness;
- break.

Armor:

- deforms;
- cracks.

Tools:

- wear out.
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12. REPAIR

Repair methods:

- blacksmith;

- craftsman;
 - self-repair;
 - magic.
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13. ITEM OWNERSHIP

Every item may have an owner.

Examples:

- player;
- NPC;
- faction;
- settlement.

Stolen items may be recognized.

14. STOLEN PROPERTY

Effects:

- guards investigate;
 - merchants refuse purchase;
 - owners seek revenge.
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15. FOOD SYSTEM

Food properties:

- nutrition;
 - freshness;
 - quality;
 - disease risk.
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16. SPOILAGE

Food spoils over time.

Examples:

Fresh ↓ Old ↓ Spoiled ↓ Rotten

17. DRINKS

Examples:

- water;
- beer;
- wine;
- spirits.

Effects:

- hydration;
 - morale;
 - intoxication.
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18. MEDICINE

Medicine properties:

- healing;
 - poison treatment;
 - disease treatment;
 - side effects.
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19. POISONS

Poisons may be:

- applied;
 - consumed;
 - inhaled.
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20. BOOKS

Books contain:

- knowledge;

- lore;
- skills;
- maps.

Books may be:

- false;
 - outdated;
 - incomplete.
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21. MAPS

Maps reveal:

- regions;
- roads;
- secrets.

Map quality varies.

22. KEYS

Keys may:

- break;
 - be copied;
 - be forged.
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23. CONTAINERS

Examples:

- bags;
- chests;
- barrels;
- crates.

Containers have:

- weight;
- volume;

- ownership.
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24. HIDDEN ITEMS

Items may be:

- concealed;
- smuggled;
- hidden.

Detection depends on:

- perception;
 - search quality.
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25. WEAPON ATTRIBUTES

Weapons possess:

- damage type;
 - weight;
 - balance;
 - reach;
 - durability.
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26. ARMOR ATTRIBUTES

Armor possesses:

- protection;
 - weight;
 - mobility penalty;
 - durability.
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27. ARTIFACTS

Artifacts may possess:

- unique abilities;

- curses;
 - history;
 - intelligence.
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28. CURSED ITEMS

Possible effects:

- addiction;
 - madness;
 - corruption;
 - power.
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29. ITEM HISTORY

Important items store history.

Examples:

- creator;
 - owners;
 - battles;
 - legends.
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30. ECONOMIC VALUE

Item value depends on:

- rarity;
 - demand;
 - location;
 - politics;
 - quality.
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31. STACKING

Only certain items stack.

Examples:

- arrows;
- coins;
- herbs.

Equipment never stacks.

32. QUICK ACCESS

Quick slots:

- weapon;
 - medicine;
 - potion;
 - torch.
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33. INVENTORY SEARCH

Players may search inventory by:

- category;
 - weight;
 - value;
 - rarity.
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34. THOUGHT SYSTEM INTEGRATION

Examples:

• My sword is damaged. • I have little food left. • This medicine may help. • Carrying this armor is exhausting.

35. NPC REACTION

NPCs react to:

- equipment;
- wealth;

- stolen items;
 - legendary artifacts.
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36. AI INTEGRATION

AI systems know:

- what the player carries;
 - what the player lacks;
 - item condition;
 - ownership.
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37. DEATH AND ITEMS

After death:

- items remain;
 - items may be stolen;
 - items may become historical artifacts.
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38. STONE SLEEP INTEGRATION

During Stone Sleep:

- food spoils;
 - items age;
 - equipment deteriorates.
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39. IMMERSION RULE

The player should feel:

"I carry possessions."

Not:

"I manage database entries."

40. CORE RULE

Items are not statistics.

Items are part of the world's history.